NYC Restaurant Health Analysis

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Health Analysis Objective

Primary Question: What are NYC's health risk hotspots, and what are the top violations?

Goal: Identify geographic risks to optimize public health inspections.

Stakeholders & Beneficiaries

DOHMH: Deploying inspection resources and scheduling inspections.

Residents: Have more transparent safety data and reduce risk of foodborne illness.

Restaurant Owners: Improving quality of food and customer experience.

Dataset & Source

Dataset Used: DOHMH New York City Restaurant Inspection Results.

Source: Obtained as a CSV file from the NYC Open Data website.

Link: DOHMH New York City Restaurant Inspection Results | NYC Open Data

What is being Analyzed?

Unit of Analysis: Each individual row in the data represents one single restaurant inspection.

Important Features:

- **Location:** Borough and Zip Code.
- **Cuisine:** The type of food served.
- **Results:** The health grade (A, B, C) and the specific violation description.

Tools Used

Jupyter Notebooks: The main workspace for writing and running the project code.

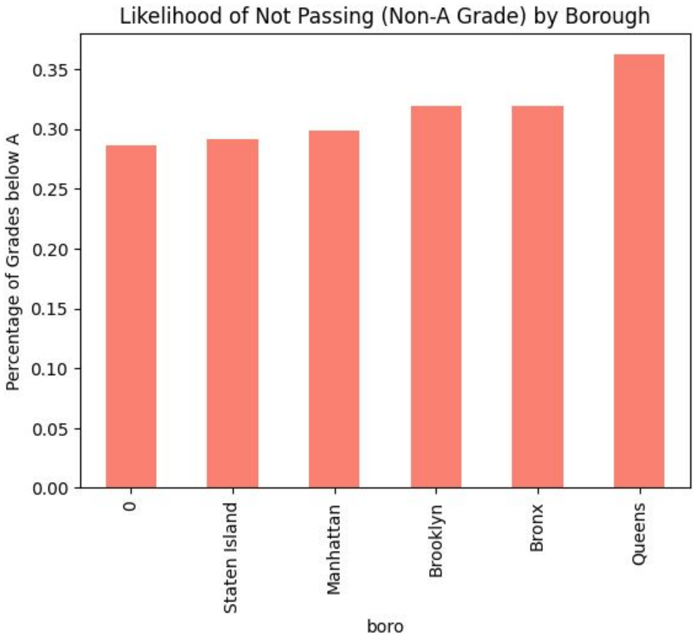
Python: The programming language used for all calculations.

Pandas: A tool used to clean the data and group it by borough or zip code.

Matplotlib: The software used to create the bear charts and pie charts.

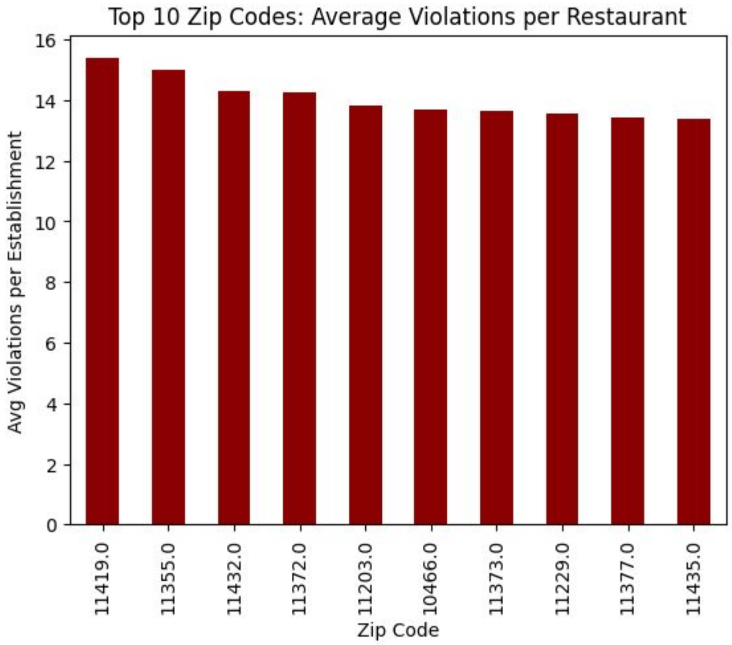
Geographic Risk by Borough

- Queens has the highest percentage of restaurants failing to earn an “A” grade, followed by the Bronx and Brooklyn.
- **Conclusion:** Geographic location is a moderate predictor of whether a restaurant will pass its inspection.



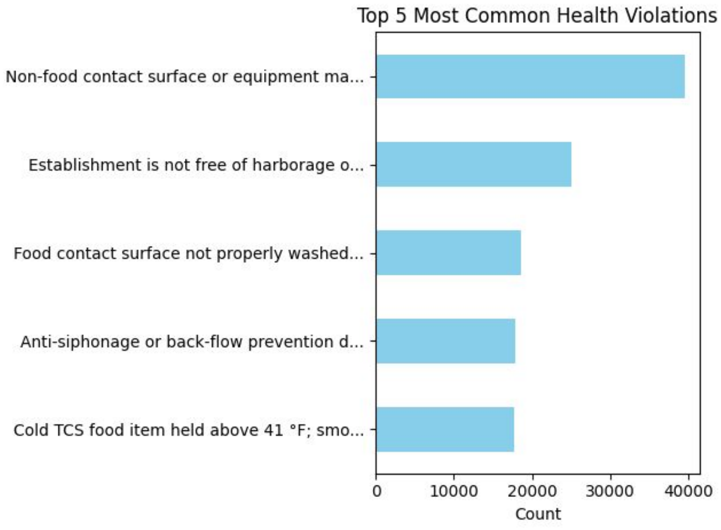
Violation Hotspots

- Zip codes 11419 and 11355 are hotspots for violations, with over 13 violations per establishment on average.
- Has many cuisines that are susceptible to health risks.
- Normalizing this data revealed specific neighborhood risk.



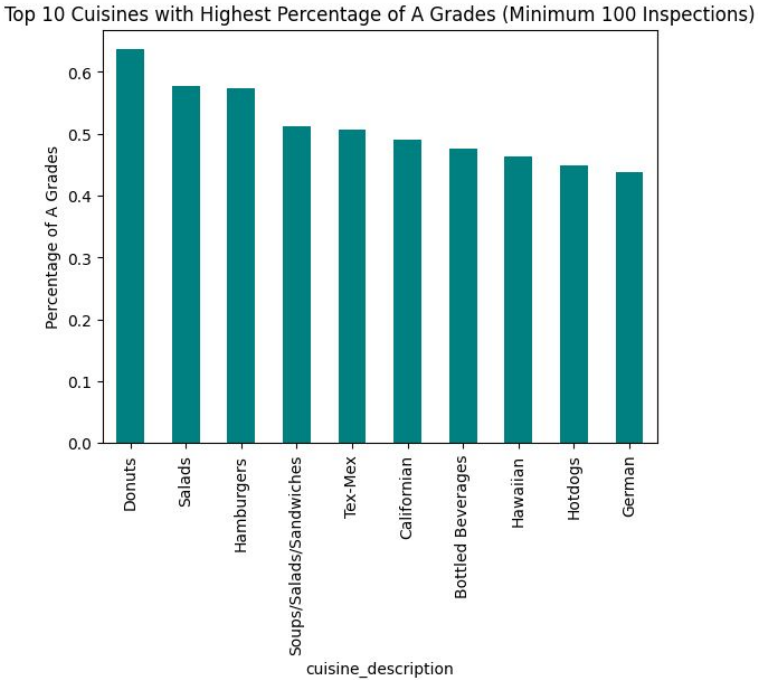
Most Common Health Violations

- The most frequent issues are vermin control and dirt non-food surfaces.
- Most restaurants struggle with facility maintenance and pest prevention rather than food safety.



Top Performing Cuisines

- Donuts, Salads, and Bottled Beverages consistently achieve “A” grades.
- Lower food-handling complexity (like beverages and salads) leads to significantly higher safety compliance.



Analysis Summary

- **Conclusion:** My analysis successfully identified which boroughs, zip codes, and violation types are the biggest risks. I was also able to gain new insights into why specific restaurant types of areas do better.
- **New Questions:**
    - Do vermin violations spike during specific seasons (like summer)?
    - Do large restaurant chains perform better than local “mom-and-pop shops”?

Future Work

- Correlated restaurant grades with neighborhood income levels.
- Create predictive models to forecast future inspection failures.
- Analyze if high inspector turnover affects consistency in grading.

Appendix

- Violation density was calculated as (Total Violations / Unique Establishments) in top 10 hazardous zip codes graph.
- **Data Source:** https://data.cityofnewyork.us/Health/DOHMH-New-York-City-Restaurant-Inspection-Results/43nn-pn8j/about_data